PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Benefits and safety of transdermal glyceryl trinitrate in acute stroke: a systematic review and meta-analysis of randomised trials (Protocol)
AUTHORS	Lim, Beng Leong; Lee, Wei Feng; Ng, Wei Ming; Tay, Wei Ling; Chan, Wui Ling

VERSION 1 – REVIEW

REVIEWER	Shinichiro Uchiyama Clinical Research Center for Medicine, International University of Health and Welfare
REVIEW RETURNED	14-Oct-2020

GENERAL COMMENTS	Scores (mRS, etc.) and timing (3 months, 6 months, 1 year, etc.) for late functional outcome status should be clearly stated. Classification of target BP lowering should also be described.

REVIEWER	Paula Munoz Venturelli Universidad del Desarrollo, Chile
REVIEW RETURNED	26-Oct-2020

GENERAL COMMENTS	This article addresses an important topic in the stroke treatment with a well written proposal, and the results of this systematic review and meta-analysis could potentially provide relevant information. In the introduction section, the authors present the aim of the study including the key elements necessary to develop a specific and well-designed review question. However, the description of the control and outcomes lack details. In the abstract, I suggest changing the introduction to focus on the limitations of the previous 2016 review to the new evidence available since last review. Methods section: The study design is appropriate to solve this research question. Regarding the search strategy, it covers several sources without language restrictions. I suggest adding a new subtitle for the outcomes. Details for the outcomes are missing. The authors should give the conceptual and operational definitions for each outcome. Will late functional outcome include death and dependency at end of follow up measured? if yes, with the modified Rankin score, other? Keywords used in the search strategy were reported; however, it is suggested to include the Mesh terms. Record management mechanism for selection and data extraction was properly described. Two pairs of independent reviewers will extract data. The approach for resolving discrepancies was stated. I suggest using a specific standardized form for data extraction,

	which should be designed and proved to minimize errors and to reduce the time required for this stage. Authors report the use of validated tools for risk of bias (RoB 2.0) and quality of evidence (GRADE) assessment. Regarding the risk of bias assessment, there is a statement for the classification of the overall level of risk of bias, which indicated only the criterion for high risk. Criteria should be defined for the other 2 categories ("some concern" and "low") and the reference should be stated. Regarding the use GRADE Tool, authors should establish which are the main outcomes that will be represented in a summary of findings table using GRADEpro. Subgroup analyses proposed by the authors are adequate, but it would be important to consider also <2 hours in the analysis of time from stroke to randomization. Regarding data analysis, on page 12, the heterogeneity cut-off for the meta-analysis should be stated. Regarding the trial sequential analysis, only two primary outcomes were considered; it is suggested to include all primary outcomes. Authors should also describe the mechanism to deal with the missing data. Regarding the references: all the references are appropriate to the review subject, but a relevant paper should be included and discussed. This article is the Cochrane SR 2017* published by the same author of SR 2016. *Bath PM, Krishnan K, Appleton JP. Nitric oxide donors (nitrates), L-arginine, or nitric oxide synthase inhibitors for acute stroke. Cochrane Database Syst Rev. 2017 Apr 21;4(4):CD000398. doi: 10.1002/14651858.CD000398.pub2. PMID: 28429459; PMCID: PMC6478181. Minor comments: Review punctuation in line 9 page 6 (;)

VERSION 1 – AUTHOR RESPONSE

Reviewer 1

1. **Comment:**
Scores (mRS, etc.) and timing (3 months, 6 months, 1 year, etc.) for late functional outcome status should be clearly stated. Classification of target BP lowering should also be described.

Authors' response:
We thank the reviewer for the invaluable comment. We have clarified the way of assessment of late functional status using mRS (with reference attached) and the BP parameters to be considered as outcome measurements in the third paragraph under the subheading "Eligibility Criteria".

Reviewer 2

1. **Comment:**

This article addresses an important topic in the stroke treatment with a well written proposal, and the results of this systematic review and meta-analysis could potentially provide relevant information. In the introduction section, the authors present the aim of the study including the

key elements necessary to develop a specific and well-designed review question. However, the description of the control and outcomes lack details. In the abstract, I suggest changing the introduction to focus on the limitations of the previous 2016 review to the new evidence available since last review.

Authors' response:

We thank the reviewer for his invaluable comment. We have added in the introduction section of the abstract to highlight that the meta-analysis in 2016 (and the Cochrane review in 2017) revealed benefits in a small patient subgroup with ultra-acute stroke and larger studies focusing on this patient subgroup have since been conducted. We have shortened bits of the introduction section of the abstract to conform with the 300 word-limit for the abstract.

2. **Comment:**

Methods section:

The study design is appropriate to solve this research question. Regarding the search strategy, it covers several sources without language restrictions. I suggest adding a new subtitle for the outcomes. Details for the outcomes are missing. The authors should give the conceptual and operational definitions for each outcome. Will late functional outcome include death and dependency at end of follow up measured? if yes, with the modified Rankin score, other?

Authors' response:

We thank the reviewer for the comment. We have expanded on our conceptual and operational definitions for the primary outcome of BP effects and late functional outcome by stating what we are examining for in terms of BP parameters and late functional outcome i.e use of modified Rankin Scale (mRS) at 3 months or later following stroke onset which is the preferred outcome for acute stroke trials (reference included). In addition, the components of the mRS are also described. These appear in the third paragraph under "Eligibility Criteria".

3. **Comment:**

Keywords used in the search strategy were reported; however, it is suggested to include the Mesh terms.

Authors' response:

We have added the MeSH terms in the section of "Search Strategy".

4. **Comment:**

Record management mechanism for selection and data extraction was properly described. Two pairs of independent reviewers will extract data. The approach for resolving discrepancies was stated. I suggest using a specific standardized form for data extraction, which should be designed and proved to minimize errors and to reduce the time required for this stage.

Authors' response:

We thank the reviewer for the invaluable comment. We have added a data extraction form adapted from the Cochrane collaboration (with reference attached) and presented as Appendix 2. The part on data collection form is added in the first paragraph of the section “Data Extraction”.

5. **Comment:**

Authors report the use of validated tools for risk of bias (RoB 2.0) and quality of evidence (GRADE) assessment. Regarding the risk of bias assessment, there is a statement for the classification of the overall level of risk of bias, which indicated only the criterion for high risk. Criteria should be defined for the other 2 categories (“some concern” and “low”) and the reference should be stated.

Authors’ response:

We thank the reviewer for the comment. We have clarified our stance regarding what constitutes low overall RoB and unclear overall RoB. This appears in the second paragraph under the section “risk of bias assessment”.

The reference for the Cochrane Collaboration’s tool for assessing RoB in randomized trials has already been stated in our study protocol (reference 25 in this revised manuscript) and that covers the various domains assessed within the tool. As for the overall RoB for each individual trial, as stated in the reference, the process requires judgement by the reviewers and they have to state their stance.

6. **Comment:**

Regarding the use GRADE Tool, authors should establish which are the main outcomes that will be represented in a summary of findings table using GRADEpro.

Authors’ response:

We thank the reviewer for the comment. We have created and attached a summary of findings table (adapted using GRADEpro); in which we will show how we will present our GRADE assessment for the main outcomes.

We have added the bits of adding a summary of findings table (adapted from Gradepro) to show how we will present our GRADE assessment for the quality of evidence for the main outcomes in the first paragraph under “Quality of Evidence”.

7. **Comment:**

Subgroup analyses proposed by the authors are adequate, but it would be important to consider also <2 hours in the analysis of time from stroke to randomization.

Authors’ response:

We thank the reviewer for his feedback. We have added the subgroup analysis of time from stroke to randomization ≤ 2 hrs vs > 2 hrs in our planned subgroup analysis. This appears in the third paragraph under the section “Data Analysis”.

8. **Comment:**

Regarding data analysis, on page 12, the heterogeneity cut-off for the meta-analysis should be stated.

Authors' response:

We thank the reviewer for the invaluable comment. When we first wrote our protocol for this proposed meta-analysis, we were reluctant to state the cut-off values for I^2 statistic for assessing heterogeneity (and decision to pool studies) as there are concerns in the literature that any cut-off values of I^2 can be misleading and the value of I^2 statistic increases with the number of patients included in each trial of the meta-analysis.

This concern of the I^2 has been described in the following article.

Rucker G, Schwarzer G, Carpenter JR, et al. Undue reliance on I^2 in assessing heterogeneity may mislead. BMC Med Res Methodol. 2008;8:79.

We feel that the decision to pool studies should be based on the clinical relevance of any heterogeneity present and the precision behind the results reported (related to sample size).

9. **Comment:**

Regarding the trial sequential analysis, only two primary outcomes were considered; it is suggested to include all primary outcomes.

Authors' response:

We thank the reviewer for the comment. We have added TSA analysis for all primary outcomes; ie in-hospital mortality, BP lowering and late functional status. We have amended this in the fifth paragraph under the section of "Data Analysis" to include the primary outcome of BP lowering and to specify that we will use mean difference of continuous outcome reported by the primary trials for TSA analysis.

10. **Comment:**

Authors should also describe the mechanism to deal with the missing data.

Authors' response:

We are grateful to the reviewer for the comment. We will address missing data in the primary studies by contacting primary authors for clarification, assessing rates of missing data and to see if intention-to-treat analysis and sensitivity analyses had been undertaken to address the impact of missing data in the primary studies.

We will judge whether these steps (or lack of them) taken by primary authors to address missing data are likely to affect or not affect their reported effect sizes.

We will run separate sensitivity analyses that include and exclude these at-risk studies (judged that missing data is likely to affect reported effect size estimates) to see if how our pooled results will change. Finally, we will also assess the risk of missing data (attrition bias) of the primary studies through our Risk of Bias and GRADE assessment.

We have added these in the fourth paragraph under the section “Data Analysis”.

11. **Comment:**

Regarding the references: all the references are appropriate to the review subject, but a relevant paper should be included and discussed. This article is the Cochrane SR 2017* published by the same author of SR 2016.
*Bath PM, Krishnan K, Appleton JP. Nitric oxide donors (nitrates), L-arginine, or nitric oxide synthase inhibitors for acute stroke. Cochrane Database Syst Rev. 2017 Apr 21;4(4):CD000398. doi: 10.1002/14651858.CD000398.pub2. PMID: 28429459; PMCID: PMC6478181.

Authors’ response:

We are grateful to the reviewer for the comment. We have included the above-mentioned 2017 Cochrane review in the Introduction of the main text (fifth paragraph) and abstract. The studies included in this Cochrane review and in the meta-analysis of 2016 published by the same authors were the same and their conclusions were similar in both reviews.

12. **Comment:**

Review punctuation in line 9 page 6 (;)

Authors’ response:

The two ; have been changed to ,.

VERSION 2 – REVIEW

REVIEWER	Shinichiro Uchiyama Clinical Research Center for Medicine, International University of Health and Welfare, Tokyo, Japan
REVIEW RETURNED	05-Dec-2020

GENERAL COMMENTS	The authors well responded to my comments.